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Security in Financial Information Systems

Blockchain - 1

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Motivation

Turkey Applies Money Laundering Rules to Crypto

Turkey added providers of cryptocurrency assets to the list of institutions that need to abide by money laundering and terrorism-financing regulations.

https://www.bloomberg.com/news/articles/2021-05-01/turkey-applies-money-laundering-rules-to-crypto-firms

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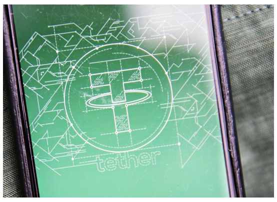
Motivation

Crypto's Shadow Currency Surges Past Deposits of Most U.S. Banks



Tether, the crypto stable-coin backed one-for-one by fiat currencies, surpassed $50 billion in circulation, a sum that's more than the insured deposits at all but 44 of the thousands of U.S. banks.

https://www.bloomberg.com/news/articles/2021-05-01/crypto-s-shadow-currency-surges-past-deposits-of-most-u-s-banks?srnd=premium-middle-east

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Motivation

Bitcoin Rises to Two-Week High After Breaking Technical Barrier

Bitcoin surged to its highest levels since mid-April after surpassing a closely-watched technical hurdle.

https://www.bloomberg.com/news/articles/2021-04-30/bitcoin-rises-to-two-week-high-after-breaking-technical-barrier?srnd=cryptocurrencies

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Motivation

Why Argentina is embracing cryptocurrency

In Argentina, there are traces everywhere of distrust and even trauma related to the economy.

https://www.bbc.com/news/business-60912789

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Outline

- Blockchain Awareness
- Introduction
- Distributed Systems
- Fundamental Concepts for Blockchain

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What is Blockchain?

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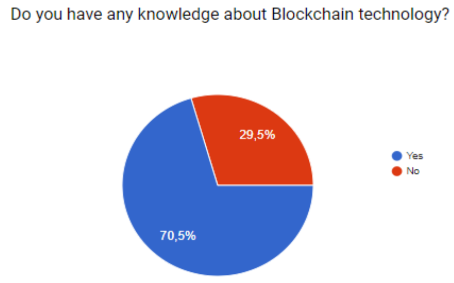
What is Blockchain?



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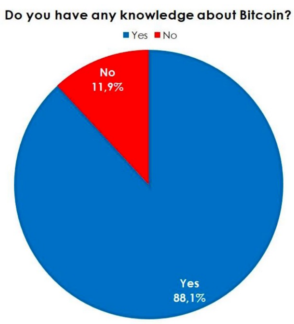
What is Blockchain?



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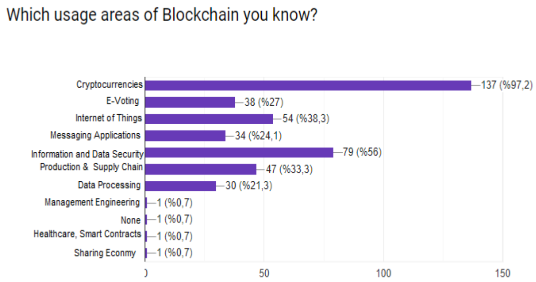
What is Blockchain?



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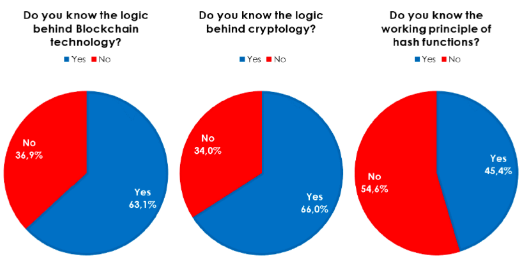
What is Blockchain?



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Introduction

The Birth of Blockchain

- Satoshi Nakamoto
- Based on bitcoin's open-source code
- Started as a non-profit
- Bitcoin: A Peer-to-Peer Electronic Cash System, 2008

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Introduction

- Cryptocurrency allows online payments to be sent directly from one party to another without going through a financial institution.

- The loss of trust in fiat currency system has brought attention to cryptocurrency.

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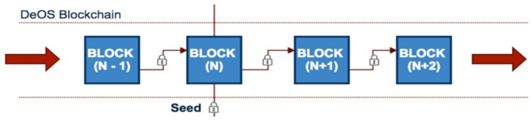
Introduction

A **Block Chain** is a distributed database that maintains a continuously growing list of ordered records called blocks.

- Records transactions
- Open
- Contains timestamp



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Introduction

Some Benefits of Blockchain

- Decentralized trust
- Cost efficiency

- Transparency
- Transaction efficiency

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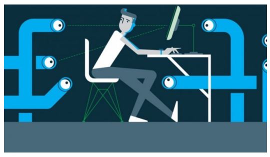
Introduction

Some Challenges of Blockchain

- Scalability

- Confidentiality
- Privacy



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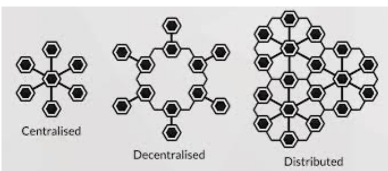
Distributed Systems

- Distributed systems are a computing paradigm whereby two or more nodes work with each other in a coordinated fashion in order to achieve a common outcome.
- Users see a distributed system as a single logical platform.



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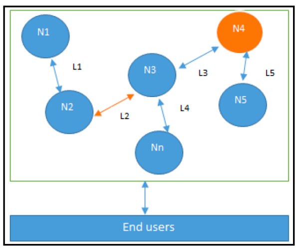
Distributed Systems

- Node is defined as an individual player in a distributed system.
- Node that can exhibit arbitrary behavior is known as Byzantine node.



Design of a distributed system: N4 is a Byzantine node, L2 is broken or a slow network link

Imran Bashir, Mastering Blockchain: Distributed ledger technology, decentralization, and smart contracts

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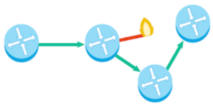
Distributed Systems

- Main challenge in distributed system design is coordination between nodes and fault tolerance.



- CAP theorem proves and states that a distributed system cannot have all properties simultaneously.

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Distributed Systems

CAP Theorem

- Consistency ensures all nodes in distributed system have a single latest copy of data
- Availability means that system is up and is accepting incoming requests and responding with data without any failures as and when required
- Partition tolerance ensures that if a group of nodes fails distributed system still continues to operate correctly

A distributed system cannot have all the three properties at the same time.

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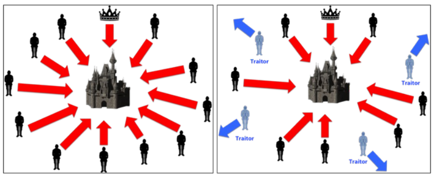
Byzantine Generals Problem

A group of army generals who are leading different parts of the Byzantine army are planning to attack or retreat from a city. The only way of communication between them is a messenger and they need to agree to attack at the same time in order to win.



Coordinated Attack Leading to Victory Uncoordinated Attack Leading to Defeat

https://medium.com/@paul_12056/byzantine-generals-problem-ff4bdc340e56

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Byzantine Generals Problem

Problem: Some generals can be traitors and can communicate a misleading message.

Solved in 1999 by Castro and Liskov, Practical Byzantine Fault Tolerance (PBFT) algorithm.

In 2009, first practical implementation was made with **bitcoin** where Proof of Work (PoW) algorithm was developed to achieve consensus.

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Consensus

Consensus is a process of agreement between distrusting nodes on a final state of data.

- Multiple nodes participating in distributed system need to agree on a single value -> very difficult to achieve consensus.
- Achieving consensus between multiple nodes is known as **distributed consensus**.

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Consensus

Consensus Requirements

- Agreement: All honest nodes decide on the same value.
- Termination: All honest nodes terminate execution of the consensus process and eventually reach a decision.
- Validity: The value agreed upon by all honest nodes must be the same as the initial value proposed by at least one honest node.
- Fault tolerant: The consensus algorithm should be able to run in the presence of faulty or malicious nodes (Byzantine nodes).
- Integrity: No node makes the decision more than once.

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Types of Consensus

- Byzantine fault tolerance-based
  - No compute intensive operations such as partial hash inversion
  - Relies on a simple scheme of nodes that are publishing signed messages
  - When a certain number of messages are received, then an agreement is reached
- Leader-based consensus mechanisms
  - Requires nodes to compete for leader-election lottery
  - The node that wins it proposes a final value

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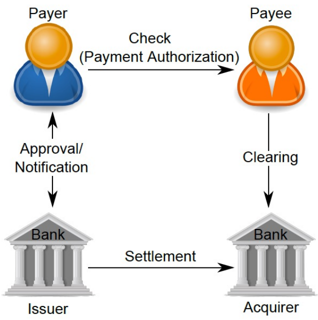
Electronic Cash



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Electronic Cash

Fundamental issues in e-cash systems

- Accountability
- Anonymity

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Electronic Cash

David Chaum in 1984 by introduced blind signatures and secret sharing.

- Blind signatures allow signing a document without actually seeing it.

- Secret sharing allows the detection of using the same e-cash token twice (double spending).

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Electronic Cash

Chaum, Fiat, and Naor (CFN) introduced anonymity and double spending detection for e-cash schemes called the concept of security reduction.

Security reduction is a technique used in cryptography to prove that a certain algorithm is secure by using another problem as a comparison.

A cryptographic algorithm is as hard to break as some other hard problem.

Can you give an example?

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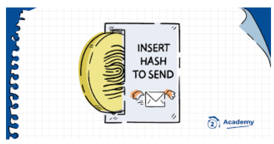
Electronic Cash

Hashcash

- Hashcash was introduced by Adam Back in 1997
- As a Proof of Work (PoW) system to control e-mail spam
- Idea: if legitimate users want to send e-mails then they are required to compute a hash as a proof that they have spent a reasonable amount of computing resources before sending the e-mail.



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Electronic Cash

Hashcash

- Hashcash takes a considerable amount of computing resources.
- It is easy and quick to verify.
- Hashcash is popularized by its use in the bitcoin **mining process**.
- Wei Dai proposed the idea of creating money via solving computational puzzles such as hashcash

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Electronic Cash

There were no clear solution of disagreements between nodes to reliance on a central trusted third party and trusted timestamping until 2009, first implementation of bitcoin.

Bitcoin solved

- The problem of distributed consensus in trustless network
- Uses public key cryptography with hashcash as PoW to provide a secure, controlled, and decentralized method of minting digital currency.
- Key innovation is the idea of an ordered list of blocks composed of transactions and cryptographically secured by the PoW mechanism.

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Summary

- Blockchain awareness
- Basics of blockchain
- Distributed systems
- Byzantine generals problem
- Consensus
- Electronic cash

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Questions?

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